

Avinash Dhiran

[Website](#) | +91 9326832776 | dhiranavinash@gmail.com | [LinkedIn : Avinash Dhiran](#)

Work Experience

- Research Intern, SARMAP SA, Caslano, Switzerland September 2026 - November 2026
- Project Intern, InfiCorridor Solutions Pvt. Ltd., Mumbai August 2024 - August 2025
 - Project Title: Analysis of SAR Data for Enhancing Land-Cover Classification using Meta-Heuristic Optimizations
 - **Research** based project, in collaboration with **ISRO**
 - Extraction and classification of **L-Band ALOS-2 PALSAR PolSAR data**
 - Employed **ACO** for feature selection and **PSO** for tuning **Random Forest**, achieving up to 12% accuracy improvement
 - Utilized of **GA** for feature set reduction, achieving up to 15% increased accuracy
 - Implemented **transfer learning** for land-cover classification
- Teaching Assistant, Department of Electronics and Telecommunications, VIT, Mumbai January 2025-April 2025
 - Course: **Digital Image Processing (Semester 6)**
 - Responsibilities: Project mentoring, Assignment designing and evaluation, Lab work
- Chanakya Fellowship Program, Mumbai April 2023-December 2024
 - Grant: 12 Lakhs INR by **Department of Science and Technology (DST), Government of India**
 - Institution: TIH Foundation for IoT & IoE (TIH-IoT), **IIT Bombay**
 - **Project Lead (2023-2024)**
 - Project Title: Design and Development of a Real-Time Multilingual Sign Language Translation Application
 - Deployed **MediaPipe** and **Fully Connected Neural Network** model for hand sign recognition with high accuracy
 - Used MediaPipe, **Spiking Convolution Neural Network** and **Bi-LSTM** model for hand gesture recognition with high accuracy
 - Designed and developed a web application using **HTML**, **CSS** and **Java Script** for real-time translation
 - Project Fellow (2022-2023)
 - Project Title: Design and Development of an Onion Harvester
 - Functionalities: Soil loosening, Onion uprooting, systematic collection

Publications

- [Real-Time Hand Gesture Recognition System Using MediaPipe, Spiking Convolution Neural Network and Bi-LSTM](#), International Conference on Electronics, Computing, Communication and Control Technology (ICECCC 2025), Bengaluru, India.
- [Real-Time Static Hand Sign Recognition System using MediaPipe and Fully Connected Neural Network](#), International Conference on Innovative Trends in Information Technology (ICITIIT 2025), Kottayam, India.
- [Polsarhub: A Large-Scale Repository for PolSAR Data](#), IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2024), Goa, India.
- [MAT SAR: A Comprehensive Machine Learning Approach for PolSAR Data Processing](#), International Journal of Computer Applications (IJCA), 2025.
- [Transfer Learning-based Approach for Polsar Image](#), International Journal of Computer Applications (IJCA), 2026.
- [Comparative Analysis of Frequency-Domain Filters for Speckle Reduction in PolSAR Imagery](#), International Journal of Computer Applications (IJCA), 2026.
- [Copyright](#), Government of India: **Computer Software for CAN Satellite** - Embedded C, December 2021

Projects

- RD-FNOSAR: A **Conditional Residual Diffusion** with Fourier Neural Operator for PolSAR De-speckling
- Land-Cover Classification of PolSAR Data using **Ant Colony Optimization** for Feature Selection and **Particle Swarm Optimization** for Hyper Tuning Random Forest's Parameters
- FNO-SAR: A **Fourier Neural Operator** for PolSAR De-speckling
- Analysis of **GAN**-Based De-speckling for PolSAR Data
- PolSAR Image Classification using **Probabilistic Adaptive Genetic Algorithm** for Feature Optimization

Education

- **Master of Science**, Faculty of Geo-Information Science and Earth Observation (ITC), **University of Twente**, Netherlands 2025-2027
 - Major: **Geo-Information Science and Earth Observation**
 - Specialization: **GEO-AI**
- **Bachelor of Engineering**, Vidyalankar Institute of Technology (VIT), **University of Mumbai**, India 2021-2025
 - Major: **Electronics and Telecommunication**
 - Minor: Artificial Intelligence and Machine Learning